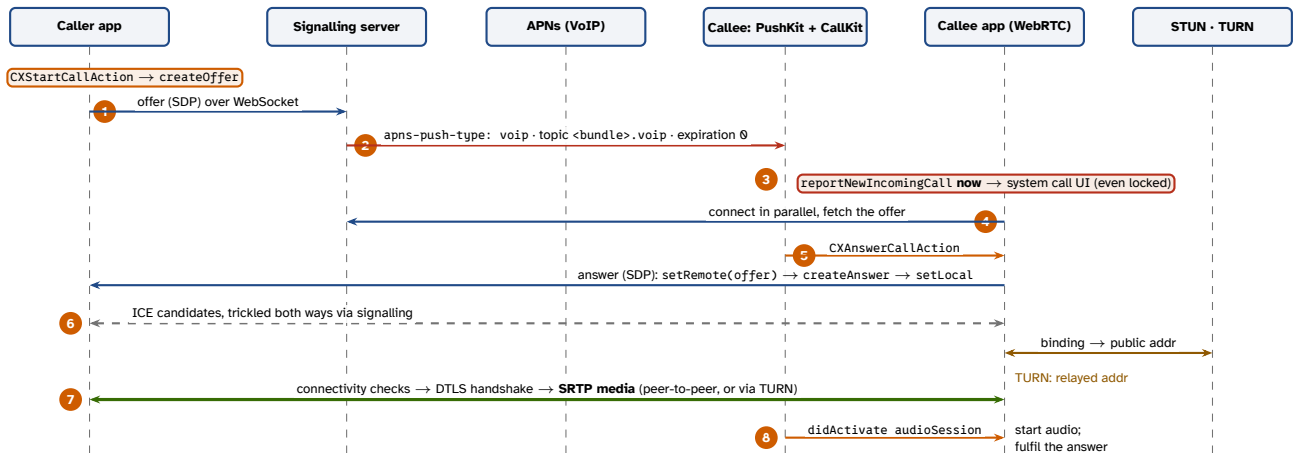


In one line: A call is two planes: **signalling** (your server, any transport — usually a WebSocket — carrying **SDP offer/answer** and **ICE candidates**) and **media** (WebRTC: **ICE** finds a path through NATs with **STUN**, falls back to a **TURN** relay, then **DTLS-SRTP** encrypts audio/video peer-to-peer). On iOS the app is usually not running when the call arrives: a **PushKit VoIP push** wakes it, it *must* report the call to **CallKit** at once, and CallKit owns the system call UI and the audio session.



WebRTC

- Signalling is not specified** by WebRTC (JSEP, RFC 8829/9429): you move SDP and candidates over your own channel. **SDP** describes codecs, media sections, ICE ufrag/pwd and the **DTLS certificate fingerprint**. Offer/answer (RFC 3264) (*unverified*): createOffer → setLocalDescription → send; the peer setRemoteDescription → createAnswer → setLocalDescription → send back.
- ICE** (RFC 8445) gathers **candidates**: *host* (LAN IP), *server-reflexive* (public IP:port a **STUN** server saw, RFC 8489), *relay* (a **TURN** server forwards, RFC 8656). Pairs are tested with STUN connectivity checks; the best working pair wins. **Trickle ICE** (RFC 8838) (*unverified*) sends candidates as they are found.
- NAT**: STUN + simultaneous sends punch holes in most NATs; a *symmetric* NAT or a UDP-blocking firewall defeats it → TURN (UDP, TCP or TLS on 443). TURN costs you bandwidth — budget for it. Network change (Wi-Fi → cellular) → **ICE restart**.
- Security**: media is always encrypted. **DTLS** handshake on the chosen path, checked against the SDP fingerprint; its keys feed **SRTP** (RFC 5764) (*unverified*). So a tampered signalling channel = MITM: protect it (TLS + auth). Data channels = SCTP over DTLS.
- Codecs**: Opus audio; VP8 and H.264 mandatory for video. **Group calls**: mesh (N–1 uplinks each: small groups only). **SFU** forwards streams (simulcast layers) · MCU mixes (server CPU).
- On iOS**: no Apple WebRTC framework for native apps — ship libwebrtc (RTCPeerConnectionFactory, RTCPeerConnection, RTCIceServer) or a vendor SDK. With CallKit set RTCAudioSession.useManualAudio = true and enable audio only in didActivate.

Example — PushKit → CallKit (the must-report path)

```

let registry = PKPushRegistry(queue: .main)
registry.delegate = self; registry.desiredPushTypes = [.voIP]
func pushRegistry(_ r: PKPushRegistry,
  didUpdate c: PKPushCredentials, for t: PKPushType) {
  api.uploadVoIPToken(c.token) } // not the alert token
func pushRegistry(_ r: PKPushRegistry,
  didReceiveIncomingPushWith p: PKPushPayload, for t: PKPushType,
  completion: @escaping () -> Void) {
  let from = p.dictionaryPayload["from"] as? String ?? "Unknown"
  let u = CXCallUpdate()
  u.remoteHandle = CXHandle(type: .generic, value: from)
  let id = calls.uuid(for: p.dictionaryPayload) // server callId to UUID
  provider.reportNewIncomingCall(with: id, update: u) { _ in completion() }
  signalling.connect() // in parallel, not before
}
func provider(_ p: CXProvider, perform a: CXAnswerCallAction) {
  Task { await call.answer(); a.fulfill() } } // fulfil once connected
  
```

Background

VoIP push launches or wakes the app (even after a force-quit). During the call the audio + voip background modes keep it running while audio flows; after hang-up it suspends as usual. The old always-connected VoIP socket (setKeepAliveTimeout) is deprecated — PushKit replaced it.

Over WebSocket, not WebRTC

SignalR (.NET: hubs, falls back WebSocket → SSE → long polling) and **Socket.IO** (Engine.IO: rooms, acks, starts with HTTP long polling then upgrades) are *protocols on top* — a plain WebSocket client can't talk to them; use their client library.

Related: websockets-realtime (signalling transport, reconnect) · push-notifications (APNs, NSE) · background-execution (modes) · avfoundation-camera-audio (categories, interruptios) · crypto-cryptokit (DTLS keys) · system-design-sdk-messenger

CallKit + PushKit — the rules

- CXProvider** (CXProviderConfiguration: supportsVideo, supportedHandleTypes, maximumCallsPerCallGroup, ringtone, icon) = *system* → app: it reports calls (reportNewIncomingCall, reportOutgoingCall(with:connectedAt:), reportCall(with:endedAt:reason:)) and delivers *actions* to its delegate. **CXCallController** = app → system: request(CXTransaction(action:)) with CXStartCallAction, CXEndCallAction, CXSetMutedCallAction, CXSetHeldCallAction. Every action: do the work, then fulfill() or fail(). providerDidReset → end everything.
- The iOS 13 rule** (Apple, PushKit docs): linking the iOS 13+ SDK, every .voIP push **must** be reported via reportNewIncomingCall; “if you fail to report a call to CallKit, the system will terminate your app”, and repeated failures may stop VoIP push delivery. So: report first, network second.
- No CallKit (e.g. a chat “ping”) → no PushKit: use ordinary pushes + an NSE. Caller hung up before you connected → still report, then reportCall(_reason: .remoteEnded); can't reach your server → .failed. Don't send more VoIP pushes to cancel — use your connection.
- Push**: apns-push-type: voip, apns-topic = bundle ID + .voip, apns-expiration: 0 (a ring that arrives late is worse than none); token from didUpdate pushCredentials — separate from the alert token.
- Outgoing**: CXCallController.request a CXStartCallAction → your delegate's perform starts signalling, fulfill() → reportOutgoingCall(with:startedConnectingAt:) → ...connectedAt: when media flows. Recents / Siri hand the app an INStartCallIntent; it then requests the same action.

Audio session during a call

Set .playAndRecord + mode .voiceChat (echo cancellation; .videoChat for video) early, but **don't activate it yourself**: CallKit activates it (with call priority) and calls provider(_didActivate:) — start audio I/O there, stop in didDeactivate. Hold / a second call arrives as CXSetHeldCallAction; mute as CXSetMutedCallAction → disable the audio track.

Traps

- ! Doing network work before reportNewIncomingCall — the app is killed on iOS 13+.
- ! Using VoIP pushes for messages “to wake the app” — not allowed; NSE instead.
- ! setActive(true) yourself / starting audio before didActivate — silent call.
- ! “STUN is enough” — some users need TURN; no TURN = calls that never connect.
- ! Thinking WebRTC defines signalling, or that SRTP keys travel in SDP (only the fingerprint does).

Remember

Push → report → connect → answer → ICE → DTLS → SRTP → didActivate.